		Master Data Science	

Work and Practical Session: Machine Learning

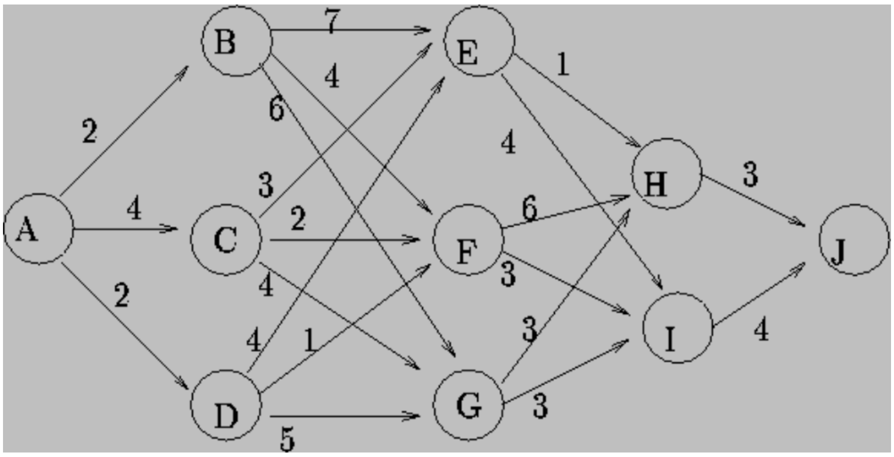
proposed by Fabrice Lauri

Session 01: Reinforcement Learning

- Work Session

Exercice #1: Graph navigation

Consider starting from node A and reaching node J in the graph below.



Define the MDP for this task, assuming that the agent wants to find the shortest route from A to J.

Exercise #2: Star gathering

Consider the following 3x3 gridworld:

*		
	T	*
	*	

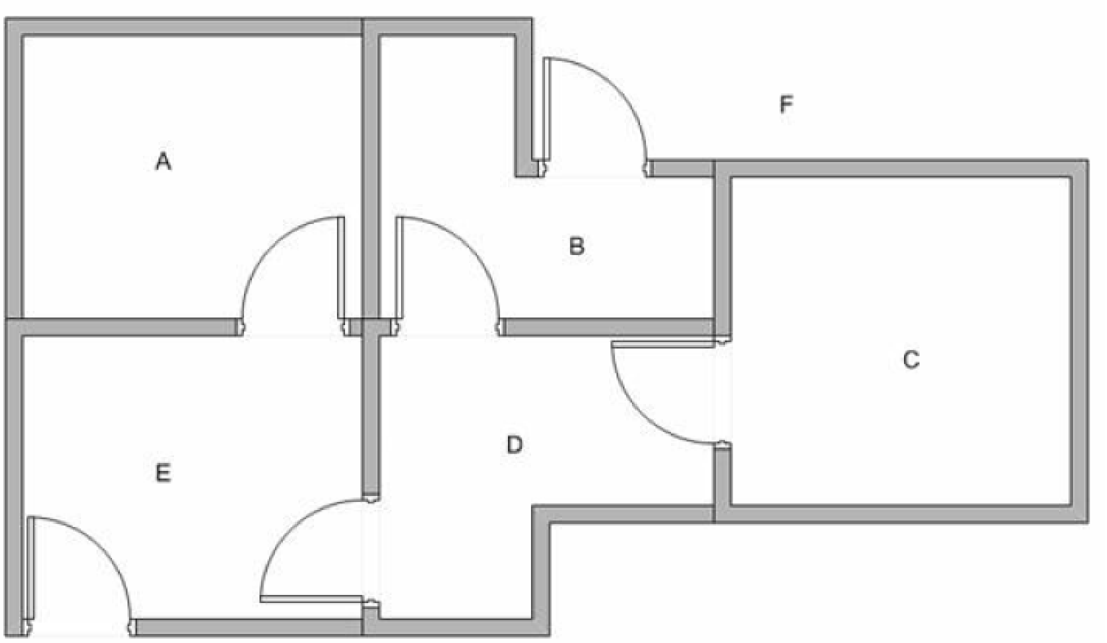
The agent's goal is to collect as many stars as possible. When the agent reaches cell T, the episode ends. The agent can move up, down, left and right. Carrying out an action that leads the agent out of the gridworld results in staying in the cell.

When an agent reaches a cell containing a star, **it receives a positive reward**. Otherwise, it receives a reward of -1. We assume $\gamma = 0.9$.

Define the MDP for this task.

Exercise #3: The rooms

Suppose we have 5 rooms in a building connected by certain doors (see the figure at the next page). We give name to each room A to E. We can consider outside of the building as one big room to cover the building, and name it as F. Notice that there are two doors leading to the building from F, that is through room B and room E.



We can represent the rooms by a graph: each room is a node and each door is an edge.

If we put an agent in any room, we want the agent to go outside of the building. In other word, the goal room is the node F. To set this kind of goal, we assign a reward to each door. The doors that lead immediately to the goal have instant reward of 100. Other doors that do not have direct connection to the target room have zero reward.

1. Formalize this problem as an MDP.
2. Apply 3 episodes of the Q-Learning, indicating all the parameters and choices used by the algorithm (starting states, transitions, with γ=0.5...) and show the final matrix of Q-values.

The Q-Learning algorithm is the next page.

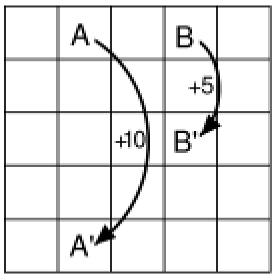
For recall, the optimality equation says: $Q*(x,u)=\rho(x,u)+\gamma \ max_{u'} Q*(f(x,u), u')$

Input: discount factor, exploration schedule $\{\Phi_k\}_{k=0}^{\infty}$
learning rate schedule $\{\alpha_k\}_{k=0}$

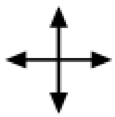
Initialize Q-function, e.g. $Q_0 \leftarrow 0$
For each episode **do**
Measure initial state x_0
for every time step k = 0, 1, 2, ... **do**
$u_k \leftarrow$ ActionSelectionMethod(x_k, Q_k ; Φ_k)
Apply u_k, measure next state x_{k+1} and reward r_{k+1}
$Q_{k+1}(x_k,u_k) \leftarrow Q_k(x_k,u_k)+\alpha_k\left[r_{k+1}+\gamma\, max_{u'} Q_k(x_{k+1},u')-Q_k(x_k,u_k)\right]$
end for
end for **Output**: $Q^* = Q_k$

Exercise #4: Stochastic policy

Consider the task of reaching cell A'. The cells of the grid correspond to the states of the environment. At each cell, four actions are possible: north, south, east, and west, which deterministically cause the agent to move one cell in the respective direction in the grid. Actions that would take the agent off the grid leave its location unchanged, but also result in a reward of -1. Other actions result in a reward of 0, except those that move the agent out of the special states A and B. From state A, all four actions yield a reward of +10 and take the agent to A'. From state B, all four actions yield a reward of +5 and take the agent to B'.



a)



Actions

3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

b)

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

c) V^*

Suppose the policy where the agent selects all four actions with equal probability in all states. Figure b shows the value functionfor this policy, for the discounted-reward case with . Figure c shows the values of the optimal policy in each state.

1. Explain why state A is valued less than 10, its immediate reward, and why the expected return of state B is more than 5, its immediate reward. For this you may use the Bellman equation.
2. Recall the Bellman optimality equation for the state value function V* of each state s. Show numerically that the Bellman equation holds for the center state, valued at 17.8, with respect to its four neighboring states. (These numbers are accurate only to one decimal place.)

From figure c, find the optimal policy

Exercise #5: Star gathering 2

Consider again the task defined in exercise #2.

1. Model this task by an oriented graph, where each node represents a state and each edge a transition between states. On each edge is indicated the action to take to go from one state to another one and the reward received from going from one state to the next one.
2. Recall the algorithm Q-Iteration.
3. Apply 3 iterations of the Q-iteration method and show the final matrix of the Q-values.
4. What is the greedy policy assuming the Q-values obtained after 3 iterations?

Exercise #6: Star collector

Consider the following 3x3 gridworld:

*		
	T	*
	*	

The agent's goal is to collect as many stars as possible, before going to the cell T. The agent can move up, down, left and right. Carrying out an action that leads the agent out of the gridworld results in staying in the cell.

Compared to exercise #2, here when the agent reaches a cell containing a star, **it removes it from the cell and receives a positive reward**. Otherwise, it receives a reward of -1. We assume $\gamma = 0.9$.

1. Describe the MDP that should be used here to solve this task, assuming there could be less or more stars at the beginning of the task and the stars could be placed at any location initially.
2. Estimate the cardinality of the state set.
3. Which algorithm(s) among Q-Iteration, Policy Iteration and Q-Learning could solve this task the fastest?

- Practical Session

The aim of this session is to program with Python the MDP of some environments and develop simple RL algorithms (Q-Iteration, Q-Learning) to automatically determine optimal action policies.

Exercise #1

Consider the environment "star gathering" of the work session #1-#2, program the MDP (using the template on Moodle), the algorithms

Q-Iteration and Q-Learning, and compare the obtained policies by varying the hyperparameters (# iterations, discount factor, learning rate…). Which behavior (action policy) do you obtain?

Exercise #2

Do some experiments with the environment "star gathering" by modifying the positive reward the agent receives when reaching a cell containing a star.

Exercise #3

We consider now the environment "star collector". In this environment, when the agent reaches a cell containing a star, it takes it.

Program it and do some experiments with this new environment.

Exercise #4

Consider the environment in problem 4 of the work session #1-#2, program the MDP (using the template on Moodle) and compare the obtained policies by using Q-Iteration and Q-Learning and by varying the hyperparameters (# iterations, discount factor, learning rate…).

Which action policy do you obtain?